Alex I.

SOFTWARE ENGINEER

Education
Computer Science and Software Engineering

Language proficiency
English – B2
Polish – B1
Swedish – A2

Domains
Embedded Systems,
Telecommunications
Library Systems
UAV Systems
Aerospace Systems

Software engineer with 14+ years of commercial experience.
I possess a strong background in both hardware and software development. My expertise mostly revolves around C/C++ software development, where I appreciate the complexity and versatility of the language, and MATLAB, where I have extensive experience in developing algorithms and applications for image analysis. I am eager to work on projects that prioritize performance and innovative features and enthusiastic about collaborating on challenging projects that allow me to leverage the full capabilities of these technologies and contribute to cutting-edge solutions.

Programming Languages
C++, C, C#, Java, JavaScript, Ruby, MATLAB, Scala.

Operating Systems
Embedded Linux.

Tools
Yocto Project, Qt Designer, Qt Designer .ui forms, System Tool Kit, systemd, CMake, qmake, gcc, GDB, Valgrind, Qt Test, Visual Studio, Qt Creator.

Frameworks & Libraries
GStreamer, Qt, Qt Widgets, QSS, Ruby on Rails, .NET, Node.js, WinForms, PCL, V4L2, Open3D, Qt DBus, Qt WebSockets.

Multimedia
WebRTC, Qt Multimedia, Linphone-SDK, liblinphone, mediasoup, libmediasoupclient.

Databases
MySQL, PostgreSQL, SQLite, MS Access.

Network
SDP, SIP, RTP, SNMP, DHCP, TFTP.

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Cloud
AWS ().

Embedded
BitBake, NXP i.MX.

Development Processes
Jira, Asana.

Source control systems
Git, GitLab, Bitbucket.

Architecture
Qt Model/View.

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Projects

SIP-BASED PHONE SOFTWARE

A SIP desk phone platform for audio and video communication, supporting both camera-enabled and audio-only device models. The project included the native phone application, web management interface, automated provisioning tools, and embedded system customization. It covered everyday calling workflows such as SIP registration, incoming and outgoing calls, video calls, call hold, transfer, conferencing, phonebook access, direct IP dialing, remote configuration, and device setup.

Project roles
Software Engineer

Period
05.2022 – till now

Responsibilities

- Refactored SIP call handling toward a cleaner MVC/MVVM-style architecture by separating UI rendering from call state, user actions, media negotiation, and Linphone-SDK integration logic;
- Authored and maintained custom BitBake recipes to cross-compile the modified Linphone-SDK and its dependencies for the NXP i.MX ARM architecture, configuring build flags to optimize memory footprint for the embedded target;
- Incrementally decomposed legacy Qt Widgets call-flow code into clearer domain, presentation, and SDK-integration boundaries while preserving existing UI behavior and QObject signal compatibility;
- Developed custom Yocto (BitBake) recipes and .bbappend files within dedicated meta-layers, enabling cross-compilation of liblinphone;
- Hardened video-call and conference UI behavior for edge cases such as held calls, second-line calls, remote hangups, stale overlays, and active video stream switching;
- Extended and customized Linphone-SDK (liblinphone) backend logic to implement SIP failover and backup-server connectivity for improved device reliability, including modifying the SDK to support authorization with different HA1 hash values for main and backup servers (necessitated by differing realm parameters);
- Developed WebRTC video-conferencing UI and signaling flows: conference creation/join, incoming invite screens, participant roster, SIP/IP/phonebook add dialogs, status strip, control panel, video tile layout, and audio/video device controls;

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- Reworked the GPIO layer built on the libgpiod C++ API (handset hook-switch edge events, RGB status LED): moved event handling to a dedicated QThread, implemented a thread-safe singleton, and fixed a mutex deadlock and a memory leak;
- Autoprovisioning tooling: migrated a WinForms-based configuration utility from .NET Framework 4.8 to .NET 8, unified project file formats across the device ecosystem, and implemented provisioning workflows using SNMP, DHCP, TFTP, and SQLite;
- Implemented on-device diagnostics: categorized logging (QLoggingCategory) into journald with size limits to protect flash storage, and a diagnostics bundle (journal, liblinphone logs, sanitized config, SIP traces) downloadable from the web interface for field issue analysis;
- Introduced Qt Test unit tests for the call-state and conference domain layer extracted during refactoring;
- Collaborated on Yocto/BitBake customization, i.MX embedded Linux target integration, and device-side service configuration for hardware-software integration.

Environment

C++17, C#, JavaScript, Embedded Linux, Yocto, Qt Designer, Qt Designer .ui forms, systemd, CMake, qmake, GCC, gdb, Valgrind, QtTest, Qt Creator, GStreamer, Qt 5.15.7, Qt Widgets, QSS, .NET 8, Node.js, WinForms, V4L2, Qt DBus, Qt WebSockets, WebRTC, Qt Multimedia, Linphone-SDK, liblinphone, mediasoup, libmediasoupclient, SQLite, SDP, SIP, RTP, SNMP, DHCP, TFTP, BitBake, NXP i.MX, Asana, Git, GitLab, Qt Model/View.

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LIBRARY SYSTEM

A robust and highly scalable library management system with over 40 years of operational history, serving a diverse client base across the globe. The system is built using a combination of C, Java, and Scala, leveraging the strengths of each language to deliver a reliable, efficient, and feature-rich platform. Designed to handle high volumes of data and complex workflows, it has continually evolved to meet the needs of modern libraries while maintaining compatibility with legacy components.

Project roles
Software Engineer

Period
05.2019 – 04.2022

Responsibilities

- Optimized and maintained the existing codebase by resolving complex bugs and implementing performance enhancements;
- Developed new backend and frontend features for socket-based communication;
- Led development efforts as a key contributor in C language, driving the successful implementation of critical features and solutions;
- Refactored existing Scala API code to improve maintainability and scalability;
- Worked with AWS pipelines, automating CI/CD processes for smoother deployments and reducing development cycle times;
- Modified database tables to support new feature implementations, improving schema design to align with updated system workflows;
- Estimated tasks with precision, ensuring realistic timelines and efficient resource allocation for development milestones;
- Conducted thorough code reviews, ensuring code quality, adherence to standards, and knowledge sharing within the basic team and client's team;
- Refactored legacy application code to improve maintainability, scalability, and overall system efficiency.

Environment
C, Java 11, Scala, gdb, MySQL, PostgreSQL, Amazon Web Services, Git, BitBucket.

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3D RECONSTRUCTION SOFTWARE FOR INDUSTRIAL APPLICATIONS

Developed cutting-edge software aimed at generating precise 3D models by processing data captured by multiple 2D laser sensors. The solution is tailored for industries requiring highly accurate 3D spatial data, such as manufacturing, robotics, and quality control.

Project roles
Software Engineer

Period
03.2019 – 05.2019

Responsibilities

- Developed software for generating 3D models from data collected by multiple 2D laser sensors, significantly enhancing accuracy in 3D reconstruction;
- Utilized the Point Cloud Library (PCL) for efficient processing and manipulation of point cloud data, optimizing data handling and performance;
- Integrated Open3D to facilitate advanced visualization and analysis of 3D models, improving user interaction and understanding of spatial data;
- Employed Visual Studio for coding, debugging, and optimizing performance, ensuring a robust and efficient software solution.

Environment
C++, Visual Studio, PCL, Open3D, Git.

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SOFTWARE MODULES FOR UAV

Developed innovative software solutions and algorithms to enhance the performance of optical sensors used in satellites and UAVs. The project included advanced photo and video processing, data analysis, and testing to improve system reliability, accuracy, and image quality for remote sensing applications.

Project roles
Software Engineer

Period
08.2017 – 12.2018

Responsibilities

- Developed software modules for digital image processing systems used in Unmanned Aerial Vehicles;
- Implemented a debayer algorithm to convert raw sensor data into full-color images, significantly improving image quality and accuracy for better visual data interpretation;
- Created image format conversion tools to support multiple formats, enhancing compatibility and integration across different systems and platforms;
- Developed an image stabilization algorithm to reduce motion-induced distortions, improving video clarity in real-time UAV operations for clearer footage capture;
- Designed an image fusion algorithm to combine data from multiple sensors, increasing detail and contrast for improved target recognition and analysis.

Environment
MATLAB, PostgreSQL, Git, BitBucket.

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REMOTE SENSING SATELLITES DATABASE

A comprehensive database of Earth remote sensing satellites, designed to store extended data for research and operational purposes. Built using Ruby and Ruby on Rails, the system provides a robust framework for managing satellite information. Data storage and querying were handled efficiently with Microsoft Access and PostgreSQL, ensuring scalability and reliability.

This database facilitated advanced analysis and streamlined workflows for satellite data management, supporting a wide range of applications in Earth observation and remote sensing.

Project roles
Software Engineer

Period
03.2016 – 03.2017

Responsibilities

- Built a web-based interface using Ruby on Rails, enabling easy access and manipulation of satellite data for users of varying technical expertise;
- Designed and optimized a PostgreSQL database to handle large datasets with detailed satellite specifications, improving data retrieval speed and enhancing user experience;
- Integrated MS Access for legacy data support, ensuring smooth data migration and compatibility with existing systems to facilitate seamless transitions.

Environment
Ruby, Ruby on Rails, PostgreSQL, MS Access.

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SOFTWARE FOR PROCESSING AND ANALYSIS OF SATELLITE EARTH OBSERVATION IMAGES

The software includes a range of modules with functions such as determining linear resolution, enhancing image quality through prediction of image type and other factors, stitching frames taken by a remote sensing satellite, eliminating optical distortions between frames. Additionally, one of the key components of the project is the development of a compression algorithm based on Adaptive differential pulse-code modulation (ADPCM).

These features are aimed at providing a comprehensive toolkit for working with satellite imagery, enabling precise resolution analysis, effective data compression, and optimized image processing workflows for various remote sensing applications.

Project roles
Software Engineer

Period
08.2012 – 03.2016

Responsibilities

- Developed algorithms for standardizing brightness levels across multiple frames and eliminating optical defects, improving the overall quality and consistency of satellite imagery;
- Designed and implemented software to analyze the linear resolution of satellite images using the sharp edge method, enhancing the accuracy of imaging performance evaluation;
- Created algorithms to assess image sharpness and edge clarity, ensuring precise resolution measurements for remote sensing applications and contributing to reliable data interpretation;
- Developed software to predict quantization ranges for remote sensing satellites by analyzing Earth observation data and estimating the albedo range, improving sensor calibration and minimizing data loss;
- Automated the prediction of quantization ranges, streamlining workflows, and improving operational efficiency;
- Designed and implemented a custom image compression algorithm using ADPCM with multiple quantizers, achieving optimized compression ratios and enhanced flexibility for different data sizes.

Environment
C++, MATLAB, System Tool Kit, CMake, Visual Studio, PostgreSQL, MS Access, Jira, Git.

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Professional skills

SKILLS		EXPERIENCE IN YEARS	LAST USED
Programming Languages	C++	8	2026
	C	3	2022
	C#	5	2026
	Java	3	2022
	JavaScript	5	2026
	Ruby	1	2017
	MATLAB	5	2018
	Scala	3	2022
Operating Systems	Embedded Linux	5	2026
Tools	Yocto Project	5	2026
	Qt Designer	5	2026
	Qt Designer .ui forms	5	2026
	System Tool Kit	4	2016
	systemd	5	2026
	CMake	8	2026
	qmake	5	2026
	gcc	5	2026
	GDB	7	2026
	Valgrind	5	2026
	Qt Test	5	2026
	Visual Studio	4	2019
	Qt Creator	5	2026
Frameworks & Libraries	GStreamer	5	2026
	Qt	5	2026
	Qt Widgets	5	2026

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SKILLS		EXPERIENCE IN YEARS	LAST USED
	QSS	5	2026
	Ruby on Rails	1	2017
	.NET	5	2026
	Node.js	5	2026
	WinForms	5	2026
	PCL	1	2019
	V4L2	5	2026
	Open3D	1	2019
	Qt DBus	5	2026
	Qt WebSockets	5	2026
Multimedia	WebRTC	5	2026
	Qt Multimedia	5	2026
	Linphone-SDK	5	2026
	liblinphone	5	2026
	mediasoup	5	2026
	libmediasoupclient	5	2026
Databases	MySQL	3	2022
	PostgreSQL	9	2022
	SQLite	5	2026
	MS Access	5	2017
Network	SDP	5	2026
	SIP	5	2026
	RTP	5	2026
	SNMP	5	2026
	DHCP	5	2026
	TFTP	5	2026

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SKILLS		EXPERIENCE IN YEARS	LAST USED
Cloud	AWS	3	2022
Embedded	BitBake	5	2026
	NXP i.MX	5	2026
Development Processes	Jira	4	2016
	Asana	5	2026
Source control systems	Git	13	2026
	GitLab	5	2026
	Bitbucket	4	2022
Architecture	Qt Model/View	5	2026

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